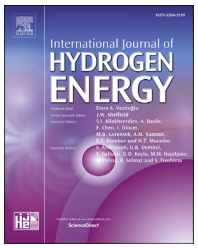
Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/he



An online prognostics-based health management strategy for fuel cell hybrid electric vehicles

Meiling Yue [a,c,*], *Zeina Al Masry* [b], *Samir Jemei* [a,c], *Noureddine Zerhouni* [b,c]

[a] *FEMTO-ST Institute, Univ. Bourgogne Franche-Comté, CNRS, Belfort, F-90000, France*
[b] *FEMTO-ST Institute, Univ. Bourgogne Franche-Comté, CNRS, Besançon, F-25000, France*
[c] *FCLAB, Univ. Bourgogne Franche-Comté, CNRS, Belfort, F-90000, France*

HIGHLIGHTS

- A health management strategy is proposed combining with on-line prognostic results.
- RUL probability is assessed using particle filtering based on particle distribution.
- Decision fusion is used to determine FLC parameters with prognostics uncertainty.
- Long-term simulation is conducted using data from real-world driving experiments.

ARTICLE INFO

Article history:
Received 2 September 2020
Received in revised form
11 January 2021
Accepted 13 January 2021
Available online 12 February 2021

Keywords:
Decision fusion
Health management
Fuel cell
Hybrid electric vehicle
Prognostics

ABSTRACT

As the energy transformation in the transportation sector is taking place driven by the development of fuel cell technologies, fuel cell hybrid electric vehicles become promising solutions owing to their long driving duration and zero emissions. However, the unsatisfied lifespan of fuel cells is an inevitable obstacle for their massive commercialization. This paper aims to propose an online adaptive prognostics-based health management strategy for fuel cell hybrid electric vehicles, which can improve the durability of the fuel cell thanks to online health monitoring. Here, particle filtering method is adapted for online fuel cell prognostics and the uncertainty of the predicted results is calculated based on the distribution of particles. A health management strategy is developed based on prognostics and a decision-making process is designed by considering the prognostics uncertainty through a decision fusion method. The obtained results show that the developed strategy has effectively improved the durability of the on-board fuel cell by up to 95.4%. Moreover, a sensitivity analysis of the prognostics occurrence frequency and probability calculation has also been conducted in this paper.

© 2021 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights reserved.

Introduction

Hydrogen, as a clean energy carrier, is leading the energy transformation in today's transportation sector and the development of fuel cell technology has accelerated this momentum. Fuel cell hybrid electric vehicles (HEVs), which offer a similar recharge time to that of the current gasoline solution and a comparable autonomy, have seen widespread use in replacing the traditional fossil fuel combustion vehicles [1]. However, on-board fuel cells are subject to unsatisfying

* *Corresponding author*. FEMTO-ST Institute, Univ. Bourgogne Franche-Comté, CNRS, Belfort, F-90000, France.
E-mail address: meiling.yue@femto-st.fr (M. Yue).
https://doi.org/10.1016/j.ijhydene.2021.01.095
0360-3199/© 2021 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights reserved.

Nomenclature

Abbreviations

EMS	energy management strategy
FLC	fuzzy logic controller
GA	genetic algorithm
HEV	hybrid electric vehicle
MF	membership function
PDM	prognostics-enabled decision-making
PEM	proton exchange membrane
PHM	prognostics and health management
SOC	state of charge
SOH	state of health
WOB	width of bin

Symbols

$D_{fc-degrade}$	fuel cell degradation ratio
i_{bat}	battery current
i_{fc}	fuel cell current
m_{H_2}	hydrogen consumption
P_{fc}	fuel cell power
Q_{max}	maximum available battery capacity
V_{bat}	battery voltage
$V_{fc-degrade}$	fuel cell degraded voltage
V_{fc}	fuel cell voltage
V_{ini}	initial voltage

durability [2]. According to the U.S. Department of Energy, the technical durability of integrated hydrogen-fueled polymer electrolyte membrane (PEM) fuel cell power systems and fuel cell stacks operating on direct hydrogen should attain 8000 h for transportation applications, however, only 4000–5000 h have been reached nowadays [3]. The challenging goal motivates the research community devoted in the development of highly efficient powertrains and specification independent control strategies that can be adopted to different hybrid fuel cell systems with various energy storage systems [4,5].

Prognostics and health management (PHM) technologies have been applied to improve the durability of fuel cells, which relies on the predictive nature of prognostics to anticipate and predict the remaining useful life (RUL) of the fuel cell and uses a post-prognostics decision support layer to design maintenance, control and management actions [6]. Prognostics approaches exist in three manners: data-driven approach, model-based approach, and hybrid approach [7]. Hybrid prognostics based on particle filtering use a set of available measurements to learn the state or the behaviour of the system and then predicts its future state. The learning could be model-based, data-driven or a combination of both because both physical models and data can be incorporated into the framework with a state vector model [8]. This method is widely used in the literature to predict the fuel cell lifetime [9,10]. However, most fuel cell prognostics research works are developed based on finished experimental degradation data, some of which are obtained under constant operating conditions. Although they are devoted to estimating the health state of the fuel cell, the real driving conditions have not been considered. Moreover, prognostics itself cannot reach the goal of improving durability, i.e., post-prognostics decisions and mitigation actions are lacking [11].

Efforts have made on the integrated design and control optimization of fuel cell HEVs in order to minimize the lifecycle cost of the vehicles. For example, a co-optimizing design and control procedure has been proposed in Ref. [5] based on an ensemble of dynamic vehicle models and heuristic control rules, which solved not only the optimal values of the installed fuel cell power and battery specific energy, but also defined the cost-oriented control policy. The strong interactions between the selected design criteria and the control strategies have enabled the researchers to exercise active control over the vehicle operations, and in turn to mitigate the deleterious effects of the fuel cell degradation [12,13]. Health-conscious energy management strategies (EMSs) have been investigated in many works by formulating multi-objective problems since the objectives of such an EMS may consist of not only minimizing the economic cost of the system but also prolonging its lifetime. The objectives are sometimes conflicting with each other so that optimal decisions need to be taken with trade-offs between the conflicting objectives. To realize the multi-objective optimization in EMSs, rule-based and optimization-based methods are used [14]. For example, fuel cell degradation origins such as low humidification and frequent and rapid voltage changes are considered in the EMS in Ref. [15] to mitigate fuel cell degradation by setting key parameters. A deterministic dynamic programming strategy and a rule-based strategy have been developed in Ref. [16] to minimize the cost and at the same time, respect the operation limits to avoid degradation. A rule-based and a model predictive controller-based power management strategies are compared in Ref. [17] and results show that the rule-based power management strategy performs better than the model predictive controller-based power management strategy under limited connected vehicle scenarios owing to the evolving dynamics between different cost factors of the fuel cell and the battery. A robust model predictive-based fuzzy control method is proposed in Ref. [18] to coordinate the fuel cell degradation and energy storage system scheduling by formulating rule-based strategies. A model predictive control strategy is proposed in Ref. [19] for the PEM fuel cell system using a linear parameter varying model, on one hand, to track the power demand, and on the other hand, to ensure the maximum working efficiency and maximizing the stack durability. Moreover, an equivalent consumption minimization strategy has been developed in Ref. [20], which considers the fuel cell degradation in the objective function. However, existing research usually considers the fuel cell degradation by setting constraints or using fitting degradation models in the strategies, which are less convincing because they are not estimated in real time. The developed fuel cell prognostics technologies are rarely applied to the energy management of HEV applications. Without an accurate understanding of the current health state, the optimization results of the EMSs may be not appropriate and may even cause more damages to the system's health, shortening its lifetime [21].

To go further from prognostics to health management for fuel cell HEV applications, Bressel et al. have proposed a multi-physical model for the PEM fuel cell in the electrochemical macroscopic representation formalism. The degradation which affects the electrochemical parameters is integrated to the control process [22]. Moreover, a prognostics-based energy

management strategy has been proposed in Ref. [23], which uses the data provided by the prognostics step to reconfigure the mission of a fuel cell HEV depending on its health state. The proposed strategy uses the median value of the predicted RULs to simply represent the current fuel cell health state. Although the strategy has improved the durability of the fuel cell system, it is less convinced if RUL uncertainties are not considered. To improve this deficiency, the uncertainties of the prognostic results are examined based on the particle filtering prognostics method in this paper. To develop the prognostics-based health management strategy, a probability calculation method is then introduced to obtain the probability values allocated to different degradation states, and a decision fusion approach is applied to calculate the confidence factors to the offline-optimized fuzzy logic controllers and to determine the parameters of the online operating controller. An optimization problem has been formulated, which determines the parameters of fuzzy logic controllers based on the power demand, the state-of-charge of the battery, the fuel consumption and the fuel cell degradation.

The reminder of the rest of this paper is as follows: Section Online prognostics based on particle filtering introduces an online prognostics method for the fuel cell based on particle filtering with an analysis of the sources of uncertainties. Section Prognostics-based health management strategy presents the development of the proposed prognostics-based EMS for a fuel cell HEV, while the simulation results are compared and discussed in Section Result discussion before concluding.

Online prognostics based on particle filtering

The online particle filtering prognostics method has been proposed in this section to be combined with the health management strategy of the studied fuel cell HEV. It allows one to track and to continuously estimate the state of health of the fuel cell, and furthermore, to predict the RUL.

Particle filtering prognostics

Particle filtering

Particle filtering is essentially a nonlinear Bayesian tracking approach that the unknown states from noisy observations can be estimated with available prior knowledge. In a typical Bayesian tracking problem, two models are required to describe the system: state model and measurement model.

State model.

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \mathbf{u}_k, \mathbf{w}_k) \leftrightarrow p(\mathbf{x}_k|\mathbf{x}_{k-1}) \tag{1}$$

where the initial distribution of the states $\mathbf{x}_k$ is modelled as $p(\mathbf{x}_0)$ and the state transition is based on a probabilistic model $p(\mathbf{x}_k|\mathbf{x}_{k-1})$. $\mathbf{u}_k$ is the command input and $\mathbf{w}_k$ is process white noise, non-necessarily Gaussian [8].

Measurement model.

$$\mathbf{z}_k = h(\mathbf{x}_{k-1}, \mathbf{v}_k) \leftrightarrow p(\mathbf{z}_k|\mathbf{x}_k) \tag{2}$$

where the measurement $\mathbf{z}_k$ is of a marginal distribution $p(\mathbf{z}_k|\mathbf{x}_k)$ and $\mathbf{v}_k$ is the measurement white noise, non-necessarily Gaussian [8].

The state estimation is realized by recursively estimating the state probability distribution function $p(\mathbf{x}_k|\mathbf{z}_{1:k})$ using a prediction step and an update step:

Prediction step.

$$p(\mathbf{x}_k|\mathbf{z}_{1:k-1}) = \int p(\mathbf{x}_k|\mathbf{x}_{k-1}) p(\mathbf{x}_{k-1}|\mathbf{z}_{1:k-1}) d\mathbf{x}_{k-1} \tag{3}$$

Update step.

$$p(\mathbf{x}_k|\mathbf{z}_{1:k}) = \frac{p(\mathbf{z}_k|\mathbf{x}_k) p(\mathbf{x}_k|\mathbf{z}_{1:k-1})}{p(\mathbf{z}_k|\mathbf{z}_{1:k-1})} \tag{4}$$

where

$$p(\mathbf{z}_k|\mathbf{z}_{1:k-1}) = \int p(\mathbf{z}_k|\mathbf{x}_k) p(\mathbf{x}_k|\mathbf{z}_{1:k-1}) d\mathbf{x}_k \tag{5}$$

In fact, the above Bayesian tracking equations are conceptual. To analytically solve this problem, particle filtering is widely applied to give approximate solutions owing to its ability to solve nonlinear and non-Gaussian problems. Particle filtering is implemented based on sequential importance sampling (SIS) technique, which is a Monte-Carlo method to calculate $p(\mathbf{x}_k|\mathbf{z}_{1:k})$ by sampling the states into particles and allocating weights to the particles. Details could be found in Ref. [24]. The steps for implementing SIS-type particle filtering are described as follows:

1. In the first step ($k = 1$), n particles with an initial distribution $p(x_0)$ are generated;
2. Propagate the n particles from x_{k-1} to the next state x_k using (1) to calculate the probability density function;
3. The weights associated to the particles are calculated by the likelihood of the state to the latest measurement $\mathbf{z}_k$:

$$\mathcal{L}(\mathbf{z}_k|\mathbf{x}_k, \sigma_{\mathbf{v}_k}) = \frac{1}{\sqrt{2\pi}\sigma_{\mathbf{v}_k}} exp(-\frac{(\mathbf{z}_k - \mathbf{x}_k)^2}{2\sigma_{\mathbf{v}_k}^2}) \tag{6}$$

4. In order to avoid the dying away of the particles, a resampling process is performed to exclude small-weighted particles by setting a given weight limit and replicate those large-weighted particles;
5. The rebuilt posterior probability density function by resampling is, therefore, recursively transferred to the following iteration as its prior probability density function;
6. Step 2 to 5 are repeated until there is no more measurement.

Prognostics require the state model to output new state estimates when the measurement is no longer available, and then to predict the future health state and the RULs. To adapt particle filtering for prognostics purpose, it needs a prediction phase in addition to the above-described state estimation process. When there is no measurement available, i.e., the prediction is started, the state $\mathbf{x}_k$ is propagated to $\mathbf{x}_{k+1}$ without calculating the likelihood of (6) and the parameters of the model are propagated with the noises.

RUL estimation

RUL is a term that indicates how long the system could remain in use, it is calculated based on a pre-defined end-of-life (EOL) threshold λ. The estimated probability density function of the RUL at time k is:

$$\hat{p}(RUL \le l|\mathbf{z}_{0:k}) = \hat{p}(\mathbf{x}_{k+l} \ge \lambda|\mathbf{z}_{0:k}) \tag{7}$$

It is updated based on new $\mathbf{z}_k$, $k = k_p + 1, \ldots, k_p + l$. The RUL uncertainty can be represented by the final distribution of the particles obtained when the failure threshold is reached [25].

The pseudo-code of implementing the particle filtering prognostics method is presented in Algorithm 1.

Algorithm 1 Particle filtering prognostics

Initialize $x_0^i, {w_0}^i, {v_0}^i, u_0^i$ with uniform distribution
Time step $k = 1$
while $x_k^i > \lambda$ and $k \le k_p$, **do**
  for $i = 1, ..., n$, **do** ▷ SIS
    Generate particles $x_k^i \sim p(x_k^i \mid x_{k-1}^i, {w_{k-1}}^i, u_{k-1}^i)$
    Allocate weight $\omega_k^i = \mathcal{L}(z_k^i \mid x_k^i, v_k^i)$
  end for
  Normalize weight $\omega_k^i = \omega_k^i / \sum_{i=1}^n \omega_k^i$
  Calculate the cumulative sum of the normalized weights $Q_k^i|_{i=1}^n = Cumsum(\omega_k^i|_{i=1}^n)$
  for $i = 1, ..., n$, **do** ▷ Resampling
    $j = 1$
    Draw a random value $u^i \sim U[0, 1)$
    while $Q_k^i < u^i$, **do**
      $j = j + 1$
    end while
    Update state $x_k^j = x_k^j$
    Update noises ${w_{k-1}}^j = {w_{k-1}}^j$, ${v_{k-1}}^j = {v_{k-1}}^j$
    Update parameters $u_k^j = u_k^j$
  end for
  $k = k + 1$
end while
for $i = 1, ..., n$, **do** ▷ Start prognostics from k_p
  while $x_k^i > \lambda$, **do**
    Propagate particles to the next time step $x_k^i = f(x_{k-1}^i, {w_{k-1}}^i, u_{k-1}^i)$
    $k = k + 1$
  end while
  Estimate $\hat{R}UL_k^i = (k - k_p) \cdot \Delta t$
end for

Online prognostics implementation

Model selection

The degradation trend of the fuel cell degradation dataset could be caught by various empirical models [25–27]. In order to include more possibilities of variation, the 2nd order exponential degradation model is chosen to track the fuel cell's performance and perform prognostics:

$$V(t) = b_1 \cdot exp(b_2 \cdot t) + b_3 \cdot exp(b_4 \cdot t) \tag{8}$$

where b_1, b_2, b_3 and b_4 are the model parameters, b_1 and b_3 determine the initial state and b_2 and b_4 define the degradation rate. Then, the state space model could be written as (9):

$$\begin{aligned} x_{1,k} &= x_{1,k-1} \cdot exp(b_2) + w_{1,k}, \quad w_{1,k} \sim \mathcal{N}(0, \sigma_{w1}^2) \\ x_{2,k} &= x_{2,k-1} \cdot exp(b_4) + w_{2,k}, \quad w_{2,k} \sim \mathcal{N}(0, \sigma_{w2}^2) \end{aligned} \tag{9}$$

where $x_{1,k}$ and $x_{2,k}$ are two first-order independent Markov processes and the present voltage state can be obtained from $x_{1,k}$ and $x_{2,k}$:

$$x_k = b_1 \cdot x_{1,k} + b_3 \cdot x_{2,k} \tag{10}$$

where b_1, b_2, b_3 and b_4 are also states to be transited. The measurement model is then written as:

$$z_k = x_k + v_k, \quad v_k \sim \mathcal{N}(0, \sigma_v^2) \tag{11}$$

where v_k are supposed to be included in the measurement signal as the measurement noise. As the initial distributions of the parameters are unknown, uniform distributions is applied [28]. The process noise variance is refined as $\sigma_w^2 = 10^{-4}$ and same process noise is assigned to the unknown parameters (b_1, b_2, b_3 and b_4). The measurement noise v_k is not considered as it is supposed to be contained in the input signals. Moreover, the number of particles is fixed as $n = 2000$.

Data preprocessing

During the system's online operation, voltage disturbances are inevitable. The original signal is too noisy to implement prognostics. Thus, the data should be preprocessed to remove the peaks and filter the noises. To this end, a kernel smoother is applied. The estimated point $f(t_j)$ is calculated based on the weighted moving average:

$$f(t_j) = \frac{\sum_{i=1}^n s_i \cdot u(t_j)}{\sum_{i=1}^n s_i} \tag{12}$$

where

$$s = K\left(\frac{t_j - t}{h}\right) \tag{13}$$

h is the bandwidth and K is a Gaussian Kernel function written as:

$$K(t) = \frac{e^{\frac{-t^2}{2}}}{\sqrt{2\pi}} \tag{14}$$

Besides, during online operation, the signal measured at each time step is too voluminous, and using these data points to estimate the states requires huge storage memory, up to gigabytes to store the states, estimated parameters, and particle weights. This would not be actually allowed in industrial applications. Besides, too many points for state estimation will lead to overfitting problems, resulting in the divergence of the model [29]. Therefore, the measured signal should be reduced. Moreover, based on the hypotheses that the degradation of the fuel cell is only due to the ageing effects and no

sudden damage happens during its operation, the degradation will not progress largely on short time scales, time scales longer than an hour should be suitable for industrial application [30,31]. Therefore, in this study, to track the state of the system and to perform prognostics, the data is reduced at one point per hour.

Prognostics result demonstration

Fig. 1 shows an example of the online estimation and prognostics result of the fuel cell degradation at prediction time $t_\lambda = 500$th hour. The fuel cell is assumed to reach its EOL at a predefined threshold ($D_{fc-degrade} = 10\%$). The predicted RUL is [84 143 284] hours with a confidence interval (CI) of [5% 50% 95%] whereas the true RUL is 159 h.

The estimated voltage drawn in Fig. 1 is the successive positions of the top of the particles distribution [25]. According to (7), the distribution of the predicted RULs is represented by the distribution of the particles when reaching failure threshold, as shown in Fig. 2.

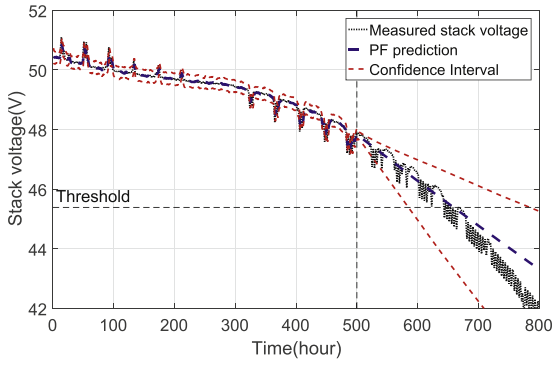


Fig. 1 – Online estimation and prognostics result of the fuel cell degradation at prediction time $t_\lambda = 500$th hour.

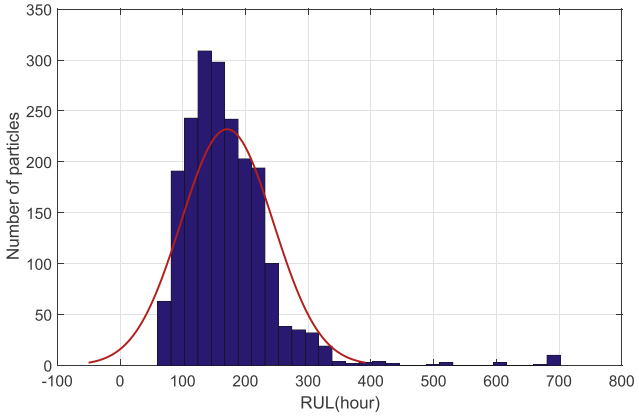


Fig. 2 – Histograms of the predicted RULs at prediction time $t_\lambda = 500$th hour and the fitted normal distribution.

Prognostics uncertainties

In fact, implementing online prognostics needs to contend with multiple sources of uncertainties. Generally, it is impossible to eliminate all the uncertainties no matter what kind of prognostics method is used. The sources of prognostics uncertainties are discussed in details:

- **Model uncertainties:** It is due to the lack of an accurate analytical model for the degrading fuel cell system. As the system is dynamic and sensible to the operation conditions, it is nonviable to find a model that covers all possible variables.
- **Input uncertainties:** It refers to the sensors that are used to monitor the system. The measurements of sensors come with a variety of noises due to electrical interference, digitization error, sensor bias, dead-band, backlash and response non-linearity, etc [32].
- **Measurement uncertainties:** All measurements are subject to uncertainties, which is a quantification of the doubt about the measurement results. Noisy measurements may due to electrical, mechanical, or even thermal condition variations.
- **External uncertainties:** It contains the uncertainties coming from the operation conditions and environment. During online operation, the operation conditions are dynamic and random and the future loading conditions and environment always remain unknown. It is another inevitable source of uncertainties.

When it comes to the particle filtering prognostics, the uncertainties of the learning phase is provided by all the above-mentioned uncertainties and can be indicated by the distribution of the particles. Then, the uncertainties of the prediction phase account for all the previously quantified uncertainties, which are used to predict the future states and their uncertainties, as well as the RULs and their uncertainties. The future states and their uncertainties are calculated by propagation using the prediction model, while the calculation of the RULs and their uncertainties depend also on the threshold function, which is used to indicate the EOL.

Prognostics-based health management strategy

A health management strategy is proposed in this section, in which the power distribution between the fuel cell and the battery is performed using fuzzy logic control (FLC), which is optimized by genetic algorithm (GA), and a prognostics-enabled decision making (PDM) process considering RUL uncertainty is designed to redistribute the power between different power sources in the studied fuel cell HEV.

System description

The studied fuel cel HEV is build in MATLAB®/Simulink® environment and the structure is shown in Fig. 3. A proportional-integral (PI) speed controller is used to track the

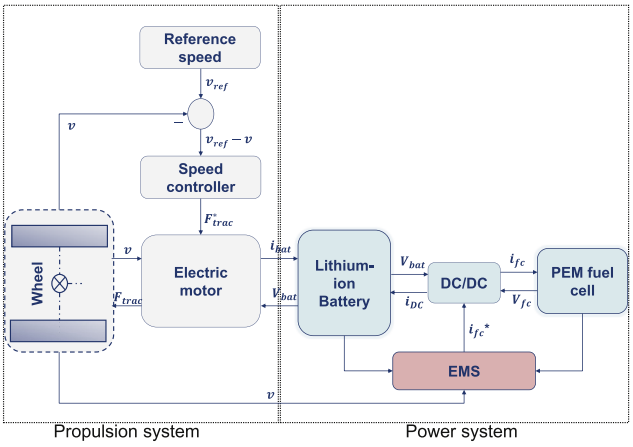


Fig. 3 – Structure of the studied fuel cell HEV.

reference speed in the propulsion system, which is modelled based on the literature [33]. The power train consists of a PEM fuel cell stack with the maximum power of 16 kW and an 80V, 40Ah lithium-ion battery pack, which are used to provide the demanded power profile. As shown in Fig. 3, the battery is connected directly to the propulsion system, while the fuel cell is connected to the battery in parallel using a boost DC/DC converter to level up its voltage. The battery current is supplied by subtracting the fuel cell current from the propulsion system. An EMS is designed to find the reference value of the fuel cell current based on the speed of the vehicle and the battery's state of charge (SOC) and then split the demanded power between the fuel cell and the battery. In order to be health-conscious, it is designed also depending on the fuel cell health state. The models of the fuel cell and the battery are detailed as follows.

Fuel cell modelling

The fuel cell in the studied vehicle is regarded as a voltage source based on the following current-controlled model, which is deduced from its polarization curve given in Ref. [33]. The fitting result is written as:

$$V_{fc} = f(i_{fc}) = \alpha_1 \cdot i_{fc}^3 + \alpha_2 \cdot i_{fc}^2 + \alpha_3 \cdot i_{fc} + \alpha_4 \tag{15}$$

The fitting factors are $\alpha_1 = -6.7791e-07$, $\alpha_2 = 0.00044927$, $\alpha_3 = -0.11913$ and $\alpha_4 = 59.124$.

The performance degradation of the fuel cell is considered on the stack level, which can be indicated by the loss of its stack voltage no matter what degradation is happening on the cell or the component level. Here, the degraded voltage of the fuel cell is modelled based on its operation points and on/off switches [34,35], written as:

$$V_{fc-degrade} = V_{ini} \cdot D_{fc-degrade} = V_{ini} \cdot (D_{fc} + D_{on/off}) \tag{16}$$

where

$$D_{fc} = \delta_0 \int (1 + \frac{\alpha (P_{fc} - P_{nom})^2}{P_{nom}^2}) dt \tag{17}$$

$$D_{on/off} = \begin{cases} \sum \Delta_{switch}, & \text{if } P_{fc,t} \geq 0 \wedge P_{fc,t-1} < 0 \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

In (17) and (18), V_{ini} is the initial available fuel cell supplied voltage, P_{nom} is the nominal fuel cell power and $P_{nom} = 6000W$, which is regarded as the maximum efficiency operation point, α and δ_0 are the load coefficients and Δ_{switch} is the voltage loss coefficient for fuel cell's once switch. The coefficients are fitted based on the historical datasets presented in Ref. [36].

Battery modelling

An electrical equivalent circuit model is used to represent the battery's behaviour, which describes the battery operating characteristics using a RC circuit network. The output battery voltage is calculated by:

$$\begin{cases} \dot{u_{C_c}} & = -\frac{u_{C_c}}{R_c C_c} + \frac{i_{bat}}{C_c} \\ V_{bat} & = u_0 - u_{C_c} - i_{bat} R_s \end{cases} \tag{19}$$

where u_0, R_s, R_c and C_c are estimated based on experimental data [33]. As there is no sensor that could read SOC directly, SOC is estimated here by ampere-hour counting (Coulomb counting) method, expressed as:

$$SOC_t = SOC_{t_0} + \eta \int_{t_0}^{t} \frac{i_\tau}{Q_{max}} d\tau \tag{20}$$

where η denotes the coulombic efficiency ($\eta = 0.95$ when charging and $\eta = 1$ when discharging), i_τ is the battery current at time instant τ ($i_\tau > 0$ when charging and $i_\tau < 0$ when discharging), and Q_{max} is the maximum available capacity of the battery.

In this study, the battery used in the fuel cell HEV is assumed to have a much longer lifetime than the fuel cell so that the battery degradation cost and the interaction between the battery and the fuel cell are not considered.

Energy management based on GA-optimized FLC

EMS is a strategy to be designed in hybrid vehicles, which defines the amount of power that must be produced by different power sources. The objectives of EMS could be consumption minimization, power source degradation mitigation, driving performance improvement, etc. Sometimes, these objectives may be conflicted with each other. In this study, the objective of the proposed EMS controller is to find a trade-off between increasing the system lifetime and saving the hydrogen consumption.

FLC design

The baseline strategy is a fuzzy rule-based strategy as it can easily be implemented in real time without huge computation burdens. The input of the FLC is the SOC of the battery and the output is the reference current of the fuel cell. The FLC is designed by considering the following objectives:

1. The range of SOC of the battery should be limited to ensure that the battery pack can provide enough power for transient power demand and at the same time, remain enough capability for regenerative braking;
2. The fuel cell is preferred to work around its highest efficiency point, P_{nom}, to save hydrogen consumption;

In this respect, the SOC of the battery is controlled to 75% as its optimal operation point and the corresponding output is the optimal i_{fc}. The control rules are written as:

1. If the SOC is *low*, then the fuel cell runs at $i_{fc-high}$;
2. If the SOC is *medium*, then the fuel cell runs at i_{fc-med};
3. If the SOC is *high*, then the fuel cell runs at i_{fc-low}.

Details of the FLC implementation could be found in the authors' previous work [23].

Offline GA optimization

Although the FLC is easy to implement in real time, it can hardly reach the optimal solution if the MFs are designed without combining any optimization procedure [11]. To encounter with this problem, in this study, the parameters of the MFs of the output i_{fc} are tuned by offline GA optimization method. GA is well adaptive to the optimization of FLCs, in which all the parameters of MFs are coded in one chromosome and they are selected, crossed and mutated to find an optimal solution to a defined objective function. Here, the objective function is a weighted polynomial function including fuel cell degraded voltage $D_{fc-degrade}$, SOC range ΔSOC and hydrogen fuel consumption m_{H_2}:

$$ObjFun = -w_1 \cdot D_{fc-degrade} - w_2 \cdot \Delta SOC - w_2 \cdot m_{H_2} \quad (21)$$

with

$$\Delta SOC = \int f\left(1 + \frac{(SOC_{bat} - SOC_{init})^2}{SOC_{init}^2}\right) dt \quad (22)$$

where w_1, w_2 and w_3 are the weighting factors. The optimization is implemented on a repeated WLTC Class 2 driving cycle and the optimized MFs under different degradation states ($D_{fc-degrade}$ is 0%, 3%, 6% and 9% of the initial stack voltage value) have been plotted in Fig. 4. The first line is the MFs of the input SOC, the second line is MFs of the optimized fuel cell reference current and the last line is the evolution of the values of the objective function (calculated by (21) and (22)).

Prognostics-based health management based on PDM

A PDM process is proposed in this section to fulfill the health management for the long term operation of the studied fuel cell HEV. A decision fusion algorithm is applied to consider the prognostics uncertainty when determining the parameters of the online FLC.

Proposed PDM process

A PDM process is to select system actions based on the predictions of future system health states [37]. Speaking about PDM in HEV applications, PDM process turns out to be a part of EMS. The proposed prognostics-based health management strategy uses the predictive nature of the prognostics process to obtain the current health state of the fuel cell and conducts energy management though FLC, which is automatically adapted to the current health state.

According to Section Online prognostics based on particle filtering, the prognostics algorithm is able to estimate the future states based on the previous degradation trend at each desired time instant. However, the degradation of power sources is a long-term procedure and it is redundant to predict their RULs on short time scales, which will lead to huge computation burdens and insufficient memory. Some researchers propose to define the time instants in order to indicate when to start performing the prognostics. According to Ref. [30], prognostics can be triggered by monitoring the health indicator. When it comes to implementing EMS, the key point is to select control actions that are suitable for the current health state. The above described health-conscious fuzzy logic control strategy tuned by offline GA optimization may be not feasible and impossible to execute in real time as the FLC cannot be tuned at each time instant and at each degradation state. Therefore, a solution is proposed in this

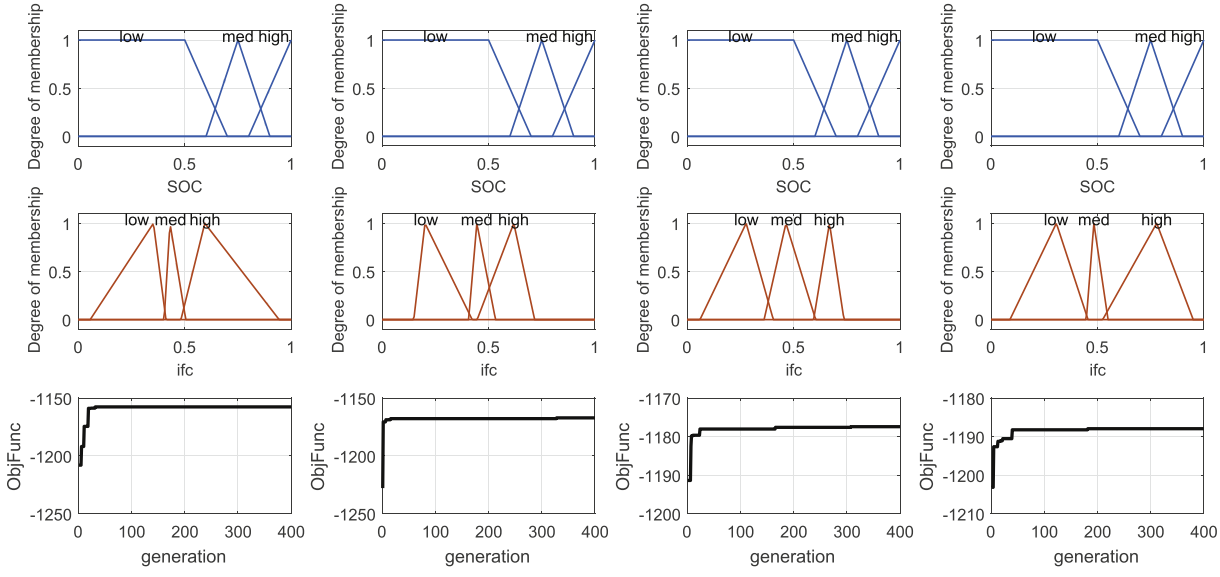


Fig. 4 – Optimized MFs and the evolution of the objective function optimized by the GA under different degradation states ($V_{fc-degrade}$ = 0%/3%/6%/9% · V_{init} from left to right).

paper, as shown in Fig. 5. The idea is to optimize a limited number of FLCs under different degradation states and when RULs are obtained through the online prognostics server, corresponding probabilities to these degradation states are calculated based on the results of prognostics. Then, a decision fusion method is applied to calculate the confidence factors to the offline optimized FLCs and refine the parameters of the online used FLC.

Probability calculation

In this study, all the pre-mentioned uncertainty factors in Section Prognostics uncertainties are assumed to follow a Gaussian distribution (that is, additive Gaussian white noise) so that the prognostics uncertainty is additive and the output RUL also follows a Gaussian distribution. It can be expressed by the distribution of particles.

The idea of considering prognostics uncertainty during the PDM process is to partition the possible solutions into bins. For example, the RUL distribution plot in Fig. 2 has been partitioned into bins with a width of approximately 20 h. Suppose that the decision to be made is to determine the probabilities of the prognostics results to a limited number of degradation cases, therefore, the number of particles indicating different RUL values can be recorded in different cases, represented by bins. By normalizing the bin counts with a sum of 1, each bin count represents the probability that an observed RUL falls within that bin, as shown in Fig. 6.

Suppose that during the online prognostics process, $RUL_{min} = 0$ h and $RUL_{max} = RUL_{CI=50\%} + t_\lambda$ hours, different widths of bins (WOB) are defined and the corresponding probability values of the RUL prediction results in Fig. 2 are calculated and listed in Table 1.

The calculation of probability is a necessary step for decision making. Online prognostics at the module level is used to assess in real time the fuel cell's health state and estimate its RUL, while during the decision-making process, the system must be auto-reconfigured depending on the health state to operate with the expected performance. The probability indicates the possibility of the fuel cell falling in a certain degradation state, which should be considered when determining the confidence factors of FLCs that have been optimized offline under certain degradation states, described in the next part of this section.

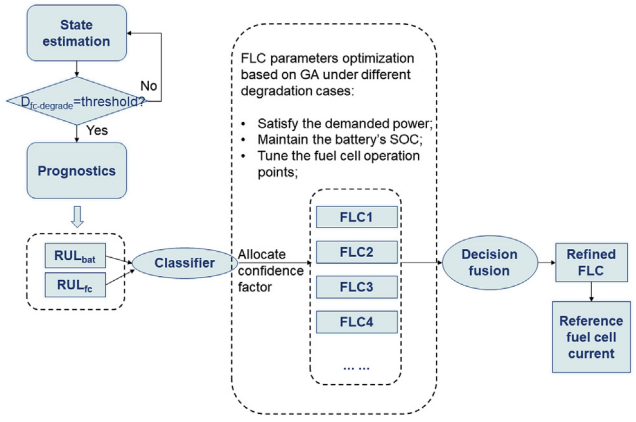


Fig. 5 – Proposed PDM structure in the prognostics-based health management strategy.

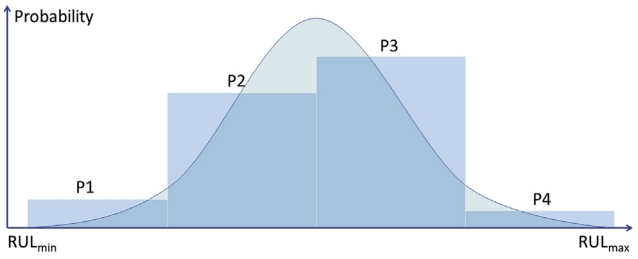


Fig. 6 – Normalizing the RUL uncertainty based on its distribution (P1+P2+P3+P4 = 0).

Decision fusion based on probability

The parameters of the online used FLC could be obtained through a decision fusion method based on probability calculation. The implementation of decision fusion is in light of the Dempster-Shafer theory. The Dempster-Shafer theory is a commonly used mathematical decision fusion method introduced by Dempster and Shafer [38,39]. Different from the Bayesian method, Dempster-Shafer theory justifies the use of combined evidence from different measures.

To combine Dempster-Shafer theory to our problem, the use of Dempster-Shafer theory to fuse offline optimization results is firstly justified. Numbering the four optimized FLCs in Fig. 4 as FLC1, FLC2, FLC3, and FLC4, one may know that FLC1 is optimized as the optimal solution when $D_{fc-degrade} = 0$, however, we don't know if it can also generate good results when $D_{fc-degrade} = 3\%$. If the range of the MFs of FLC1 contains the range of the MFs of FLC2, it means that FLC1 can work well even when $D_{fc-degrade} = 3\%$. To include this situation, when generating the online used FLC, we need to consider not only the FLC parameters for each degradation state but also the relationship between the different FLCs.

Using the basic probability assignments calculated from the online prognostics process, one can combine the evidence from different degradation states and calculate the belief measure value that takes into account all possible states according to Dempster-Shafer theory:

$$Bel(A) = \sum_{B \subseteq A} m(B) \tag{23}$$

To understand this equation, it is done in a geometrical way. As seen in Fig. 7, they are the MFs of "i_{fc} = high" extracted from Fig. 4.

To calculate the belief measure value of each side, one should longitudinally compare the range of the four S1 and four S2 of the MFs. As indicated by (23), a combined belief measure value is obtained if the range of one side is the subset of the range of another side. For example, S2 of the MF of "i_{fc} = high" in FLC1 has a larger range "a" that contains the range of S2 in FLC2 and FLC3, "b" and "c", respectively, so that the belief measure value of it is determined as P1+P2+P3. To be noted, as $x_{43} > x_{13}$, the range "a" does not contain range "d", therefore, the probability P4 is not taken into account. Other belief measure values for each side of each MFs are obtained using the same method.

Then, to obtain the basic belief assignment $m(A)$ for the MF of one case, a Möbius transformation is used given the belief measure value of each side [40]:

Table 1 – Probability calculation results with different definitions of WOB.

$WOB = 0.5 \cdot RULmax$	P1							P2		
	0.9879							0.0121		
$WOB = 0.4 \cdot RULmax$	P1				P2				P3	
	0.3208				0.6727				0.0065	
$WOB = 0.3 \cdot RULmax$	P1				P2			P3		P4
	0				0.9407			0.0558		0.0035
$WOB = 0.2 \cdot RULmax$	P1		P2		P3		P4		P5	
	0.3208		0.6199		0.0528		0.0050		0.0015	
$WOB = 0.1 \cdot RULmax$	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
	0	0.3208	0.4158	0.2041	0.0473	0.0055	0.0030	0.0020	0.0010	0.0005

$$m(A) = \sum_{B \subseteq A} (-1)^{|A-B|} Bel(B) \tag{24}$$

This equation is simplified as (25) by neglecting the term $(-1)^{|A-B|}$. This is because the cardinality of $A - B$ is uncountable and insignificant [41].

$$m(A) = \sum_{B \subseteq A} Bel(B) \tag{25}$$

As an MF consists of two sides, the assigned probability to the MF is calculated by the weighted arithmetic mean of the side belief measure values [23]. Then, the parameters of the online used FLC could be determined and adapted to the current health state.

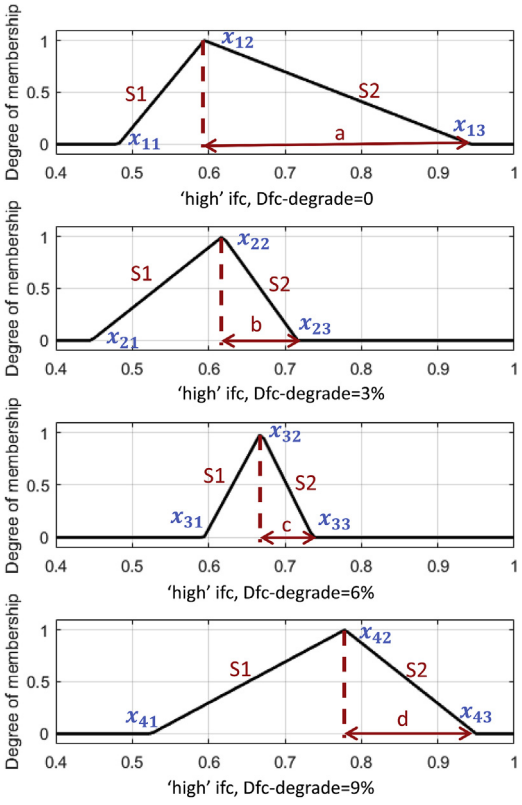


Fig. 7 – Optimized 'high' i_{fc} MFs under different degradation cases and the calculation of the belief measure value of S2 (As the range "a" contains the range "b" and the range "c", in the first MF, *Bel* (S2) = P1 + P2 + P3.).

Result discussion

Comparison analysis

The health management strategy is coded in the EMS module of the simulated fuel cell HEV model (Fig. 3). The simulation is run on a repeated WLTC class 2 driving cycle up to thousands of hours. At first, two scenarios have been proposed to justify the effectiveness of the proposed strategy. In the first test, the FLC optimized under no degradation state ($D_{fc-degrade} = 0\%$) is used all along with the simulation until the fuel cell reaches its EOL. Then in the second test, the parameters of the online used FLC are refined according to the prognostics results considering uncertainty. The WOB is defined as $0.3 \cdot RUL_{max}$ and the prognostics is triggered once $D_{fc-degrade}$ falls by 1%. The simulation stops when the fuel cell reaches the same EOL value. The evolution of the power distribution is plotted in Fig. 8, in which the fuel cell operates smoothly and the battery provides all the dynamics in the demanded power.

However, as the fuel cell degrades, the developed EMS based on optimized FLC without considering degradation will lose its optimality. The fuel cell can no longer operate with the least dynamics, as shown in Fig. 9. Large variations could be observed in the fuel cell power from the 200th hour, which contributes to its further degradation.

The proposed prognostics-based health management strategy in this paper has conducted to encounter this problem. The health state of the fuel cell is considered in the EMS through online prognostics and the FLC parameters are adapted online. The demanded power can, therefore, be redistributed between the fuel cell and the battery to avoid

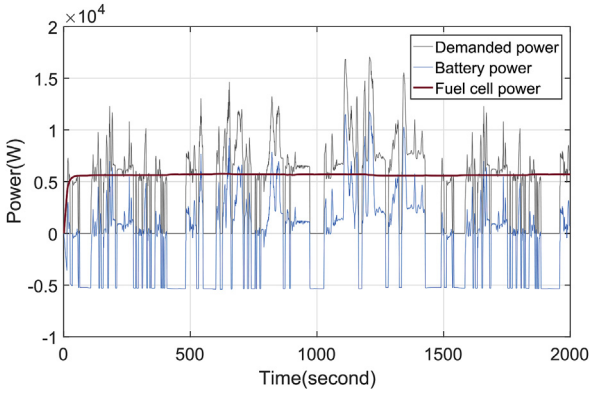


Fig. 8 – Power distribution between the fuel cell and the battery.

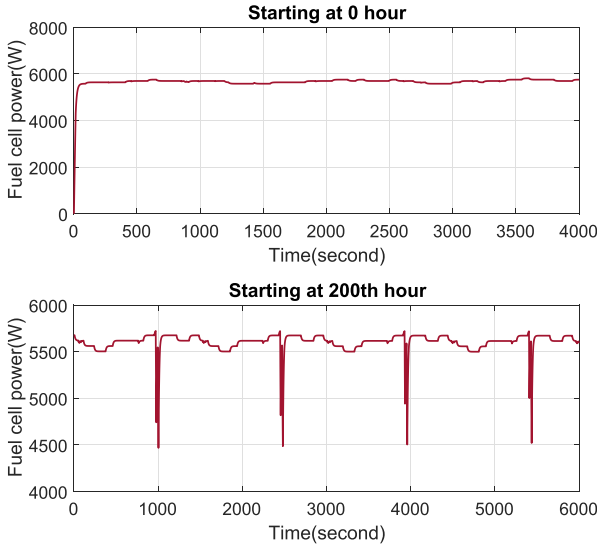


Fig. 9 – Fuel cell power at 0 h and 200th hour using EMS without prognostics.

large variations in the fuel cell provided power, as shown in Fig. 10.

To compare the durability, the evolution of the fuel cell stack voltage and the battery SOC are compared in Fig. 11 and Fig. 12. The fuel cell has reached its EOL at 763 h using the EMS without prognostics, while the prognostics-based EMS has prolonged the fuel cell lifetime to 957 h, by 25.5% approximately. Besides, compared to prognostics-based EMS, the EMS without tuning the FLC parameters cannot maintain the SOC of the battery. Battery SOC has dropped to a lower value along with the time, which also contributes to battery degradation, as shown in Fig. 12.

Discussion on prognostics occurrence frequency

To further investigate the influence of prognostics occurrence, more scenarios with prognostics happening at different

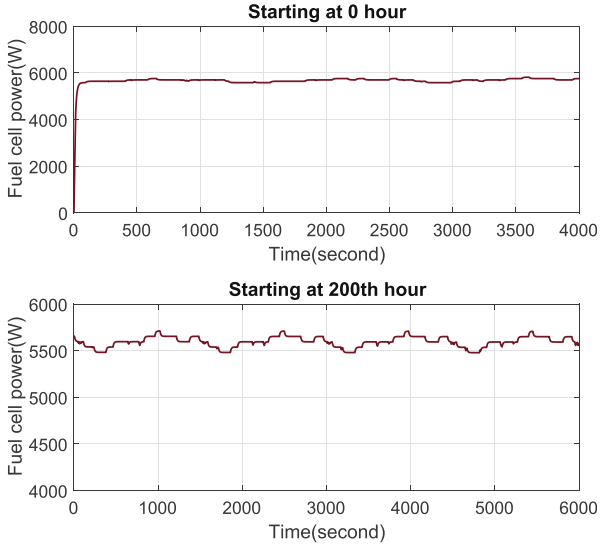


Fig. 10 – Fuel cell power at 0 h and 200th hour using prognostics-based health management strategy.

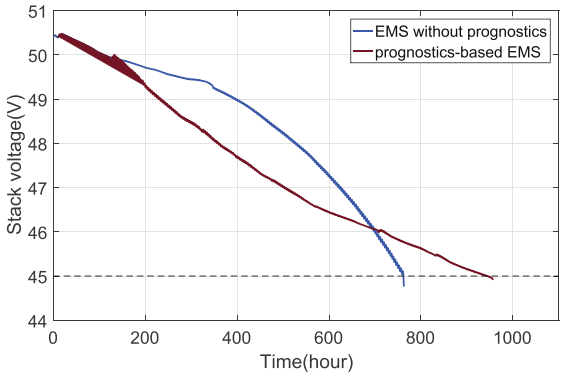


Fig. 11 – Comparison of the measured fuel cell stack voltage between the EMS non-adopting prognostics and the proposed prognostics-based EMS.

frequencies are designed, while the WOB remains at the same. The prognostics process is triggered by monitoring the fuel cell degradation with the degradation interval $\Delta D_{fc-degrade}$ equal to 1%, 2%, 3%, 4% and 5%, while the *WOB* remaining as $0.3 \cdot RUL_{max}$.

The comparison of life improvement is listed in Table 2. To demonstrate the expected benefits on the total cost of ownership, the fuel cell is assumed to cost 100 US dollars per kW based on 2020 automotive fuel cell system target defined by the US Department of Energy [42]. The cost per hour due to degradation is calculated when the fuel cell reaches its EOL. The hydrogen cost is 3.5 USD/kg based on the 2020′ projection [42]. Compared to the EMS without prognostics, the proposed prognostics-based health management strategy improves the lifetime of the fuel cell with a minimum value of 25.5% and the strategy with $\Delta D_{fc} = 3\%$ has reached the longest lifetime, 84.6% longer than the EMS without prognostics. Obviously, it doesn't mean that frequent prognostics can lead to better performance of the fuel cell durability. Frequent prognostics ($\Delta D_{fc} = 1\%, 2\%$) have even worse performance than less frequent prognostics ($\Delta D_{fc} = 3\%, 4\%, 5\%$). This is because when the prognostics horizon is large, the prognostics accuracy cannot be guaranteed due to the lack of learning data. When the uncertainty accumulates, the probability calculation will also be influenced so that one cannot reach the optimal parameters of the online used FLC.

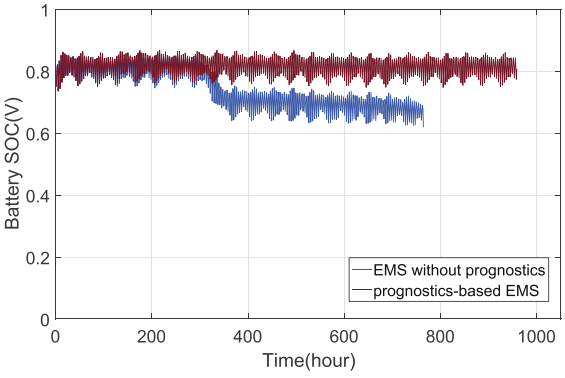


Fig. 12 – Comparison of the measured battery SOC between the EMS non-adopting prognostics and the proposed prognostics-based EMS.

Table 2 – Fuel cell lifetime and system cost comparison under different prognostics occurrence frequency.

Prognostics frequency (WOB = 0.3 • RUL_{max})	Lifetime improvement	Degradation cost (USD/hour)	m_{H_2} (g/hour)	Hydrogen cost (USD/hour)	Total cost (USD/hour)
EMS without prognostics	–	0.79	41.21	0.14	0.93
ΔDfc = 1%	25.5%	0.63	42.02	0.15	0.78
ΔDfc = 2%	58.2%	0.50	41.95	0.15	0.65
ΔDfc = 3%	84.6%	0.43	41.91	0.15	0.58
ΔDfc = 4%	80.0%	0.44	41.92	0.15	0.59
ΔDfc = 5%	54.7%	0.51	41.71	0.15	0.66

Table 3 – Fuel cell lifetime and system cost comparison under different WOBs.

WOB (ΔDfc = 3%)	Lifetime improvement	Degradation cost (USD/hour)	m_{H_2} (g/hour)	Hydrogen cost (USD/hour)	Total cost (USD/hour)
EMS without prognostics	–	0.79	41.21	0.14	0.93
$WOB = 0.1 \cdot RUL_{max}$	61.7%	0.49	41.82	0.15	0.64
$WOB = 0.2 \cdot RUL_{max}$	60.8%	0.49	41.81	0.15	0.64
$WOB = 0.3 \cdot RUL_{max}$	87.3%	0.42	41.95	0.15	0.57
$WOB = 0.4 \cdot RUL_{max}$	95.4%	0.40	41.94	0.15	0.55
$WOB = 0.5 \cdot RUL_{max}$	19.5%	0.66	41.64	0.15	0.81

However, when the frequency becomes lower, i.e., fewer prognostics are implemented, the performance becomes worse, as well. This is due to the reason that the degradation of the fuel cell hasn't been fully considered in the strategy. If we compare the hourly hydrogen consumption, the result does not vary a lot as our objective function is defined to find a trade-off between degradation mitigation and saving consumption. The EMS without prognostics consumes less because the SOC of the battery drops to a lower value. The total cost per hour for the fuel cell system including degradation cost and hydrogen cost has been reduced, which shows the expected benefits by deploying the proposed health management strategy.

Discussion on WOB of probability calculation

Further analysis of how the WOBs of probability calculation will have an influence on the system performance is discussed by defining different WOBs. More simulations are executed with $WOB = 0.5 \cdot RUL_{max}$, $0.4 \cdot RUL_{max}$, $0.3 \cdot RUL_{max}$, $0.2 \cdot RUL_{max}$ and $0.1 \cdot RUL_{max}$, respectively. The prognostics process is triggered with the fuel cell degradation interval $\Delta D_{fc-degrade}$ equal to 3%, as it has been proved to have the best performance, presented in Section Discussion on prognostics occurrence frequency.

The comparison of life improvement is listed in Table 3. It shows that the strategy with WOB of probability calculation that equals $0.4 \cdot RUL_{max}$ has reached the best performance. The durability of the fuel cell has been improved by 95.4%. However, the EMS with WOB of 0.5 can only improve the lifetime by 19.5%. This is because the FLC formulated based on the probability calculation with $WOB = 0.5 \cdot RUL_{max}$ cannot take into account the high degradation state. Besides, the EMS with smaller WOBs ($WOB = 0.1 \cdot RUL_{max}$, $WOB = 0.2 \cdot RUL_{max}$) are less satisfied due to the uncertainty in the prognostics results. Once the uncertainty is accumulated to a high level, the refined FLC cannot generate the best results. The hourly consumed hydrogen, as well as the hourly cost are compared, which shows that the proposed strategy improves the fuel cell lifetime and reduced the total cost of the ownership.

According to the above analysis, when selecting the parameters, i.e. the prognostics occurrence frequency and the WOB of probability calculation, to define the online prognostics-based health management strategy, it is important to improve the prediction performance of the prognostics method and to define the necessary points for reconfiguration according to the characteristics of the system.

Conclusion

This paper has contributed to developing an enhanced online prognostics-based health management strategy considering prognostics uncertainty for fuel cell HEVs. The RUL probability is assessed using particle filtering based on particle distribution and a decision fusion method is developed in the PDM process to determine FLC parameters with prognostics uncertainty. Long-term simulations are conducted using data from real-world driving experiments. Results showed that the proposed prognostics-based health management strategy has effectively stabilized the fuel cell operation and maintained the battery SOC, which helped to mitigate the degradation of the fuel cell without sacrificing the hydrogen consumption. Moreover, variations on the prognostics occurrence frequency and the calculation of probability have also been examined, in which the strategy with $\Delta D_{fc} = 3\%$ has reached the longest lifetime, 84.6% longer than the EMS without prognostics and the strategy with WOB of probability calculation that equals $0.4 \cdot RUL_{max}$ has reached the best performance with the fuel cell durability improved by 95.4%. It concludes that proper parameters of the strategy could help to improve the EMS performance. For the perspectives, further research on

hybrid system dynamic degradation monitoring and experimental validation are expected in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work has been supported by the French regional project PHyTie [grant number 2016Y-04574]. This work has also been supported by the EIPHI Graduate School [contract ANR-17-EURE-0002] and the Region Bourgogne Franche-Comté, France.

REFERENCES

[1] Ajanovic A, Haas R. Prospects and impediments for hydrogen and fuel cell vehicles in the transport sector. Int J Hydrogen Energy 2021;46:10049–58. https://doi.org/10.1016/j.ijhydene.2020.03.122.

[2] Fu Z, Zhu L, Tao F, Si P, Sun L. Optimization based energy management strategy for fuel cell/battery/ultracapacitor hybrid vehicle considering fuel economy and fuel cell lifespan. Int J Hydrogen Energy 2020;45(15):8875–86. https://doi.org/10.1016/j.ijhydene.2020.01.017.

[3] Kurtz J, Sprik S, Saur G, Onorato S. Fuel cell electric vehicle durability and fuel cell performance. Golden, CO: National Renewable Energy Laboratory; 2019. NREL/TP-5400-73011, https://www.nrel.gov/docs/fy19osti/73011.pdf.

[4] Feng Y, Dong Z. Integrated design and control optimization of fuel cell hybrid mining truck with minimized lifecycle cost. Appl Energy 2020;270:115164. https://doi.org/10.1016/j.apenergy.2020.115164.

[5] Sorrentino M, Cirillo V, Nappi L. Development of flexible procedures for co-optimizing design and control of fuel cell hybrid vehicles. Energy Convers Manag 2019;185:537–51. https://doi.org/10.1016/j.enconman.2019.02.009.

[6] Jouin M, Gouriveau R, Hissel D, Marion-Péra M-C, Zerhouni N. Prognostics and health management of PEMFC - state of the art and remaining challenges. Int J Hydrogen Energy 2013;38:15307–17. https://doi.org/10.1016/j.ijhydene.2013.09.051.

[7] Liu H, Chen J, Hissel D, Lu J, Hou M, Shao Z. Prognostics methods and degradation indexes of proton exchange membrane fuel cells: a review. Renew Sustain Energy Rev 2020;123:109721. https://doi.org/10.1016/j.rser.2020.109721.

[8] Jouin M, Gouriveau R, Hissel D, Péra M-C, Zerhouni N. Particle filter-based prognostics: review, discussion and perspectives. Mech Syst Signal Process 2016;72–73:2–31. https://doi.org/10.1016/j.ymssp.2015.11.008.

[9] Cheng Y, Zerhouni N, Lu C. A hybrid remaining useful life prognostic method for proton exchange membrane fuel cell. Int J Hydrogen Energy 2018;43(27):12314–27. https://doi.org/10.1016/j.ijhydene.2018.04.160.

[10] Chen J, Zhou D, Lyu C, Lu C. A novel health indicator for PEMFC state of health estimation and remaining useful life prediction. Int J Hydrogen Energy 2017;42(31):20230–8. https://doi.org/10.1016/j.ijhydene.2017.05.241.

[11] Yue M, Jemei S, Gouriveau R, Zerhouni N. Review on health-conscious energy management strategies for fuel cell hybrid electric vehicles: degradation models and strategies. Int J Hydrogen Energy 2019;44(13):6844–61. https://doi.org/10.1016/j.ijhydene.2019.01.190.

[12] Jahromi MM, Heidary H. Durability and economics investigations on triple stack configuration and its power management strategy for fuel cell vehicles. Int J Hydrogen Energy 2021;46(7):5740–55. https://doi.org/10.1016/j.ijhydene.2020.11.103.

[13] Teng T, Zhang X, Dong H, Xue Q. A comprehensive review of energy management optimization strategies for fuel cell passenger vehicle. Int J Hydrogen Energy 2020;45(39):20293–303. https://doi.org/10.1016/j.ijhydene.2019.12.202.

[14] Shen D, Lim C-C, Shi P. Robust fuzzy model predictive control for energy management systems in fuel cell vehicles. Contr Eng Pract 2020;98:104364. https://doi.org/10.1016/j.conengprac.2020.104364.

[15] Zhang Z, Miyajima R, Inada T, Miyagi D, Tsuda M. Novel energy management method for suppressing fuel cell degradation in hydrogen and electric hybrid energy storage systems compensating renewable energy fluctuations. Int J Hydrogen Energy 2018;43(14):6879–86. https://doi.org/10.1016/j.ijhydene.2018.02.124.

[16] Kandidayeni M, Macias A, Boulon L, Kelouwani S. Investigating the impact of ageing and thermal management of a fuel cell system on energy management strategies. Appl Energy 2020;274:115293. https://doi.org/10.1016/j.apenergy.2020.115293.

[17] Wang Y, Advani SG, Prasad AK. A comparison of rule-based and model predictive controller-based power management strategies for fuel cell/battery hybrid vehicles considering degradation. Int J Hydrogen Energy 2020;45(58):33948–56. https://doi.org/10.1016/j.ijhydene.2020.09.030.

[18] Shen D, Lim CC, Shi P. Fuzzy model based control for energy management and optimization in fuel cell vehicles. IEEE Trans Veh Technol 2020:1. https://doi.org/10.1109/TVT.2020.3034454.

[19] Goshtasbi A, Ersal T. Degradation-conscious control for enhanced lifetime of automotive polymer electrolyte membrane fuel cells. J Power Sources 2020;457:227996. https://doi.org/10.1016/j.jpowsour.2020.227996.

[20] Li H, Ravey A, N'Diaye A, Djerdir A. Online adaptive equivalent consumption minimization strategy for fuel cell hybrid electric vehicle considering power sources degradation. Energy Convers Manag 2019;192:133–49. https://doi.org/10.1016/j.enconman.2019.03.090.

[21] Han J, Han J, Yu S. Investigation of FCVs durability under driving cycles using a model-based approach. Journal of Energy Storage 2020;27:101169. https://doi.org/10.1016/j.est.2019.101169.

[22] Bressel M, Hilairet M, Hissel D, Ould Bouamama B. Model-based aging tolerant control with power loss prediction of proton exchange membrane fuel cell. Int J Hydrogen Energy 2020;45(19):11242–54. https://doi.org/10.1016/j.ijhydene.2018.11.219.

[23] Yue M, Jemei S, Zerhouni N. Health-conscious energy management for fuel cell hybrid electric vehicles based on prognostics-enabled decision-making. IEEE Trans Veh

Technol 2019;68(12):11483–91. https://doi.org/10.1109/TVT.2019.2937130.
[24] Arulampalam MS, Maskell S, Gordon N, Clapp T. A tutorial on particle filters for online nonlinear/non-Gaussian bayesian tracking. IEEE Trans Signal Process 2002;50(2):174–88. https://doi.org/10.1109/78.978374.
[25] Jouin M, Gouriveau R, Hissel D, Péra M-C, Zerhouni N. Prognostics of PEM fuel cell in a particle filtering framework. Int J Hydrogen Energy 2014;39(1):481–94. https://doi.org/10.1016/j.ijhydene.2013.10.054.
[26] Zhang X, Pisu P. An unscented kalman filter based approach for the health-monitoring and prognostics of a polymer electrolyte membrane fuel cell. Annual conference of the prognostics and health management society. 2012. p. 1–9.
[27] Jouin M, Gouriveau R, Hissel D, Zerhouni N, Marion-Péra M-C. Prognostics of proton exchange membrane fuel cell stack in a particle filtering framework including characterization disturbances and voltage recovery. In: 2014 international conference on prognostics and health management, PHM 2014; 2014. https://doi.org/10.1109/ICPHM.2014.7036363.
[28] An D, Choi J-H, Kim N. A tutorial for model-based prognostics algorithms based on matlab code. In: Proceedings of the annual conference of the prognostics and health management society 2012, PHM 2012; 2012. p. 224–32.
[29] Jouin M, Gouriveau R, Hissel D, Péra MC, Zerhouni N. Combined predictions for prognostics and predictive control of transportation PEMFC. IFAC-PapersOnLine 2016;49(28):244–9. https://doi.org/10.1016/j.ifacol.2016.11.042.
[30] Jouin M, Bressel M, Morando S, Gouriveau R, Hissel D, Péra M-C, Zerhouni N, Jemei S, Hilairet M, Bouamama BO. Estimating the end-of-life of PEM fuel cells: guidelines and metrics. Appl Energy 2016;177:87–97. https://doi.org/10.1016/j.apenergy.2016.05.076.
[31] Saxena A, Celaya J, Balaban E, Goebel K, Saha B, Saha S, Schwabacher M. Metrics for evaluating performance of prognostic techniques. In: 2008 international conference on prognostics and health management; 2008. p. 1–17. https://doi.org/10.1109/PHM.2008.4711436.
[32] Saha B, Goebel K. Uncertainty management for diagnostics and prognostics of batteries using bayesian techniques. In: 2008 IEEE aerospace conference; 2008. p. 1–8. https://doi.org/10.1109/AERO.2008.4526631.
[33] Depature C, Jemei S, Boulon L, Bouscayrol A, Marx N, Morando S, Castaings A. IEEE VTS motor vehicles challenge 2017 - energy management of a fuel cell/battery vehicle. In: 2016 IEEE vehicle power and propulsion conference (VPPC); 2016. p. 1–6. https://doi.org/10.1109/VPPC.2016.7791701.
[34] Chen H, Pei P, Song M. Lifetime prediction and the economic lifetime of proton exchange membrane fuel cells. Appl Energy 2015;142:154–63. https://doi.org/10.1016/j.apenergy.2014.12.062.
[35] Fletcher T, Thring R, Watkinson M. An energy management strategy to concurrently optimise fuel consumption and pem fuel cell lifetime in a hybrid vehicle. Int J Hydrogen Energy 2016;41:21503–15. https://doi.org/10.1016/j.ijhydene.2016.08.157.
[36] Gouriveau R, Hilairet M, Hissel D, Jemei S, Jouin M, Lechartier E, Morando S, Pahon E, Pera M, Zerhouni N. IEEE PHM 2014 data challenge: outline, experiments, scoring of results, winners, IEEE 2014 PHM Challenge; 2014. p. 1–6.
[37] Balaban E, Alonso JJ. A modeling framework for prognostic decision making and its application to uav mission planning. In: Annual conference of the prognostics and health management society; 2013. p. 1–12.
[38] Dempster AP. A generalization of bayesian inference. J Roy Stat Soc B 1968;30(2):205–32.
[39] Shafer G. A mathematical theory of evidence, vol. 42. Princeton university press; 1976.
[40] Liu L, Yager RR. Classic works of the dempster-shafer theory of belief functions: an introduction. Berlin, Heidelberg: Springer Berlin Heidelberg; 2008. p. 1–34. https://doi.org/10.1007/978-3-540-44792-4-1.
[41] Zhou D, Al-Durra A, Gao F, Ravey A, Matraji I, Simões MG. Online energy management strategy of fuel cell hybrid electric vehicles based on data fusion approach. J Power Sources 2017;366:278–91. https://doi.org/10.1016/j.jpowsour.2017.08.107.
[42] US Department of Energy. The department of energy hydrogen and fuel cells program plan. 2011. https://www.energy.gov/eere/vehicles/downloads/us-drive-fuel-cell-technical-team-roadmap.html.